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RESEARCH ARTICLE

Flight Test Results of an Adaptive RF Front-End for Multi-Band SDR in Avionics

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ABSTRACT This paper presents an in-depth evaluation of the reconfigurable and agile RF front-end (RFFE) architecture, previously demonstrated in laboratory settings through real-world flight tests. The study aims to validate the practical performance, reliability, and robustness of the RFFE architecture, developed in conjunction with software-defined radios (SDRs) by the LASSENA lab, in dynamic aviation environments. By transitioning from controlled lab conditions to actual flight scenarios, we assess the ability of the architecture to adapt to varying signal requirements, frequencies, and protocols. Preparation for flight situations includes temperature resilience, vibration tolerance, and mobility tests. An initial airport ground test will be conducted to ensure the system's readiness before actual flight deployment. Key performance metrics such as spectrum utilization, signal integrity, receiver performance, and transmitter linearity are examined to ensure compliance with stringent aviation safety and performance standards. The results provide critical insights into the operational benefits and potential enhancements of the RFFE architecture, supporting its adoption in aviation and potentially other fields requiring high communication and system efficiency levels.

INDEX TERMS Architecture, aviation, communication, CNS, flight test, FPGA, low noise amplifier, power amplifier, RFFE, SDR, avionics.

I. INTRODUCTION

Avionics systems depend heavily on robust and flexible communication, navigation, and surveillance (CNS) capabilities to ensure safety and efficiency. Central to these systems is the RFFE, which interfaces with antennas, receivers, and transmitters to enable wireless communication. SDRs have significantly advanced RF and wireless communication technologies, offering enhanced flexibility and performance. SDRs are particularly utilized in integrated modular avionics and satellite communication systems for high-throughput data links across various orbits. These systems leverage a flexible radio architecture with baseband algorithms implemented in field programmable gate arrays (FPGA) or digital signal processors (DSP), providing modularity, expandability, and the ability to update processing algorithms in space. This reduces costs and minimizes size by integrating most processing functions into a single chip. Despite their modular design, current SDRs are often tailored to specific applications due to unique RF and analog-digital conversion requirements. The previous study [?] introduced a reconfigurable and agile RFFE architecture that seamlessly integrates with SDR, enhancing spectrum utilization, signal integrity, and overall system efficiency [2] in laboratory settings.

Multi-band SDR systems, such as the Frontier Radio developed by JHU/APL [3], play a crucial role in enhancing communication in avionics by offering versatility and efficiency across various wireless communication standards [4]. These SDRs can support multiple frequency bands, enabling seamless integration with different avionics systems and applications, including Wireless Avionics Intra-Communication Systems (WAICSs) [5]. The flexibility of SDR technology allows for reconfigurability in flight, accommodating different modulation formats, bit rates, and coding schemes, which is essential for adapting to the

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dynamic communication requirements in aviation scenarios. Additionally, the ability of SDRs to handle concurrent multiple standards on one platform enhances communication reliability and coverage, making them ideal for applications like aircraft navigation and Air Traffic Control (ATC) networks [6].

The integration of multi-band SDR systems in avionics offers several benefits. Firstly, it enables the sharing of computational and hardware resources among different avionic radio functions, reducing size, weight, power, and cost (SWAP-C) requirements [7], [8]. Additionally, the flexibility of SDR allows for dynamic reconfiguration of resource allocation based on active radio sets, ensuring adaptability to changing communication needs and standards in the aviation industry [8]. Moreover, the use of SDR technology facilitates the implementation of wideband radio (WBR) for In-flight connectivity (IFC) services, supporting real-time data exchange between aircraft and ground stations via various links [8]. Furthermore, the ability of SDR to operate in multiple frequency bands enhances the performance and efficiency of avionics systems, offering increased bandwidth and improved communication capabilities [9].

Adaptive RFFE plays a crucial role in signal reception by dynamically adjusting RFFE circuitry to achieve a selected frequency response based on the detected RF power [10]. In the context of a Cognitive radio (CR) receiver supporting IEEE 802.22 standard, the adaptive modulation techniques significantly influence the design of the RF receiver front-end, impacting characteristics such as noise figure (NF), linearity, gain, and output power [11]. Additionally, in the mobile phone RFFE antenna combination, the performance directly affects the quality of the wireless link between the handset and cellular network base stations, underscoring the importance of adaptability in optimizing signal reception [12]. Furthermore, in the context of an adaptive Ultra High-Frequency Radio-Frequency Identification (UHF RFID) analog front end, the adaptive architecture enhances the communication range between the tag and reader, ensuring effective communication at both close and extended ranges [13].

Flight tests of an adaptive RFFE for multi-band SDR in avionics have shown promising results. The WBR module, integrated into the Multi-Mode Software Defined Avionic Radio (MM-SDAR), demonstrated successful in-flight performance with a transmission power of 10 W and a bandwidth of 675 kHz, achieving a maximum slant range of 5 NM and maintaining an average throughput of around 360 kbit/s [14]. Additionally, the MM-SDAR prototype, part of the AVIO-505 project, underwent flight tests over the past three years, confirming its operation in flight conditions and showcasing its potential benefits compared to traditional avionics systems. These tests validated the MM-SDAR's performance and highlighted areas for further development in the evolving field of avionics technology. [15]

Several challenges were encountered during avionics testing, including the need for closed-loop testing in systems like Interval Management (IM) avionics to evaluate performance accurately due to interactions with aircraft and environmental factors [?]. Additionally, integrating new technologies, such as high-power DC supplies, into existing systems like the electronic consolidated support system (eCASS), posed hardware and software challenges, requiring power supply paralleling and automated reconfiguration to overcome limitations and ensure safe operation [17]. Moreover, the development of more autonomous aircraft and green aviation technologies increased the workload for avionic infrastructure, necessitating the use of geographically distributed testing to enhance efficiency and flexibility in the testing process [18]. These challenges underscore the complexity and critical nature of avionics testing in ensuring the safety and performance of aircraft systems.

This work aims to move from controlled test settings to real-world flight tests, validating the RFFE's performance, reliability, and robustness in dynamic aviation conditions. Unlike our prior work, which focused primarily on system architecture and laboratory validation [1], this paper demonstrates the practical deployment of the SDR-based RFFE in actual flight scenarios, including important modifications to the original system. This includes meeting stringent in-flight constraints on inter-band leakage and system mechanical stability, optimizing for SWaP-C requirements, and validating performance through structured flight procedures. We further compare measured results with GPS-based references, a standard benchmarking method in avionics research, thereby offering not just functional demonstration but also quantitative performance validation. As such, this work provides an important step toward real-world readiness of adaptive SDR RFFE systems in aviation. This paper is structured as follows: Section II details the architecture design of the integrated RFFE and SDR system. It also describes flight preparation and the integration techniques that make the design practical for flight tests. Section III presents the flight test results and discusses the accuracy of the tested radio results in comparison with reference measurements, and Section IV provides the conclusion of the study.

II. ARCHITECTURE DESIGN OF THE INTEGRATED RFFE

Building upon the architecture detailed previously [1], the current system incorporates critical enhancements specifically designed for real-world avionics deployment. These improvements, motivated by regulatory constraints and operational necessities identified during extensive field preparations, ensure that the architecture meets stringent aviation standards for SWaP-C, robustness, and inter-band leakage performance.

This section provides a comprehensive overview of the implemented design, including the RF pathway and associated components. The block diagram in Figure 1 illustrates a multi-band RF transceiver system tailored for aviation communication and navigation, amalgamating various critical functionalities such as Very high frequency (VHF)

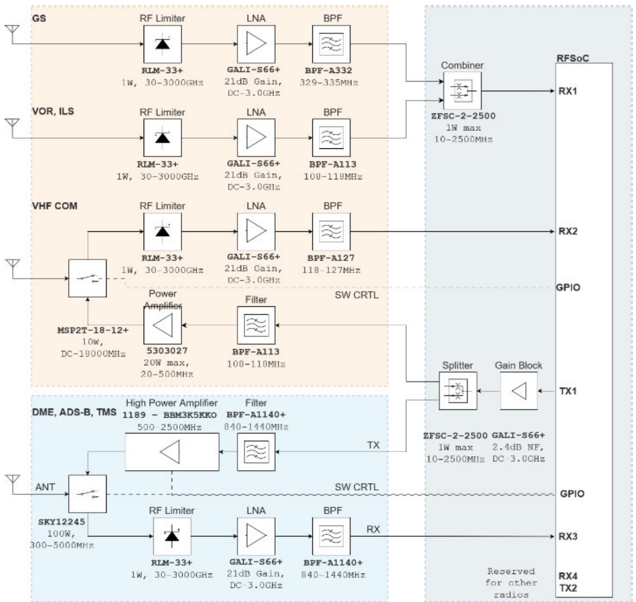

FIGURE 1. RFFE architecture.

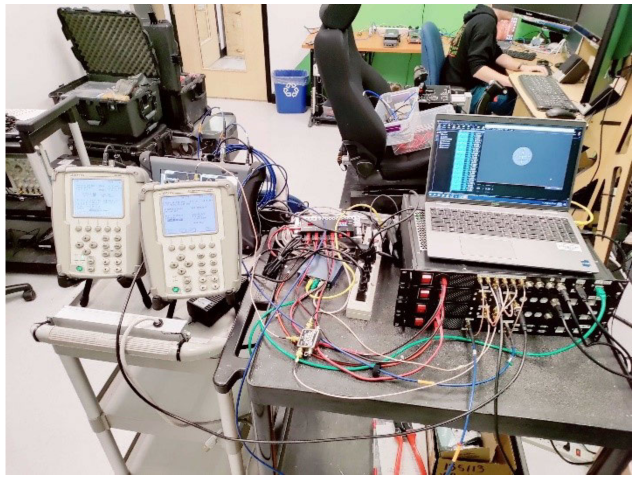

FIGURE 2. SDR and RFFE test setup in the lab.

Omni Directional radio (VOR), Instrument landing system (ILS), Glide slope (GS), Very high-frequency communication (VHF COM), Automatic Distance measuring equipment (DME), Dependent surveillance-broadcast (ADS-B), and Transponder mode-s (TMS). Each section within the diagram is meticulously designed to ensure seamless operation across different frequency bands, fostering reliability and robustness in aviation operations.

The RF pathway is meticulously structured, comprising essential components like RF limiters, low noise amplifiers (LNA), bandpass filters (BPF), mixers, power amplifiers (PA), and RF switches, each serves a distinct purpose in signal processing. Whether amplifying weak signals with minimal noise, filtering out unwanted frequencies, or converting signal frequencies for further processing, every component plays a vital role in maintaining signal integrity and fidelity.

The system is adept at both receiving and transmitting signals, with dedicated paths for each function. Incoming signals from antennas undergo a series of amplification and filtering stages to prepare them for processing within the RFSoC. Conversely, outgoing signals are meticulously processed to ensure compliance with regulatory standards and are transmitted precisely via antennas.

This integrated approach not only optimizes the performance of individual communication and navigation systems but also enhances overall system efficiency and reliability, making it an ideal choice for aviation applications where seamless communication and precise navigation are paramount. Figure 2 shows the whole system implemented in separated rack boxes with connections. Each box has a sort of RF connectors, power supply ports, and signal handling connections. IFR6000 is used to simulate DME, TMS, and ADS-B radio signals and IFR4000 is used for VOR, ILS, GS, and VHF COM signal simulation. The use of this equipment helped the team verify both the RFFE and the SDR code before going to the next step.

A. SDR DATA SAMPLING

Receiving RF signals on the RFSoC 4×2 SDR is a critical process for high-performance applications such as wireless communications, radar systems, and spectrum analysis. The RFSoC 4×2 combines RF signal conversion and digital processing in a single chip, enabling direct sampling of RF signals up to 4 GHz. This direct RF sampling capability reduces the need for multiple intermediate frequency stages, thus minimizing latency and preserving signal integrity. The key techniques for receiving RF signals on the RFSoC 4×2 include advanced digital down-conversion, which allows for precise tuning and filtering of specific frequency bands. This process involves mixing the received RF signal with a local oscillator signal to convert it to a lower intermediate frequency or baseband, followed by digital filtering to isolate the desired signal. Multi-channel reception is another important feature of the RFSoC 4×2, allowing simultaneous reception and processing of multiple RF signals. This capability is particularly beneficial in applications like Multiple Input Multiple Output (MIMO) systems, where multiple spatially separated antennas improve communication performance. Synchronization and calibration are also essential to ensure accurate signal representation and minimize errors. This involves aligning the phase and timing of the received signals across different channels and correcting any amplitude and phase imbalances.

The FPGA on the RFSoC 4×2 also plays a crucial role in the overall system functionality. It supports custom digital signal processing algorithms to enhance flexibility and performance and generates control signals for RF switches and other time-sensitive components. These control signals are vital for managing the signal path and ensuring precise timing and coordination of various elements within the RF receiver system. The RFSoC 4×2 SDR provides a robust and versatile platform for advanced RF signal reception and processing. It integrates these advanced techniques and leverages the powerful FPGA capabilities.

B. FLIGHT TEST PREPARATION

1) RECEIVER BLOCK REFINEMENT

In the first iteration, the new receiver block, which integrates an LNA and a BPF, was designed and tested in the lab, demonstrating excellent performance. This initial design prioritized achieving high performance and facilitating ease of testing within a controlled laboratory environment. Building on these results, the subsequent design iteration maintained high-performance standards while emphasizing ease of setup and operation during flight. Figure 3 illustrates how the receiver block evolved to meet real-world constraints, particularly those encountered during flight tests. The updated design also focused on improving the receiver block's resilience to vibration and electromagnetic interference, which are critical factors in aviation environments. Additionally, efforts were made to minimize the overall size of the receiver block to comply with the stringent space and weight limitations typical of avionics systems.

FIGURE 3. Receiver block with metal shield.

The Noise Figure of the L-band receiver block, as illustrated in Figure 4, shows the noise figure plot obtained from the Agilent N9030A signal analyzer coupled with noise source model 346C. At 1.0897 GHz, the system exhibits a 3.81 dB noise figure. This value is a critical metric for receiver performance, as it represents the signal-to-noise ratio (SNR) ratio at the input to the SNR at the output. A noise figure of 3.81 dB indicates that the receiver introduces noise power equivalent to half of the thermal noise at the input. In practice, a noise figure below 3 dB is considered optimal for sensitive communication systems, but this noise figure is deemed sufficient for general-purpose receivers, especially in wideband systems.

The measurement results in Figure 4 also reveal two significant noise spikes near 876.7 MHz and 889.8 MHz, where the noise figure reaches 14.92 dB and 14.23 dB, respectively. These elevated values can be attributed to leakage from GSM-850 and GSM-900 Downlink signals, as the measurements were not conducted in an anechoic chamber. The presence of strong external signals in the vicinity of the receiver's operating band likely contributed to these spikes by introducing interference into the system, which was captured as additional noise. Despite this interference, the noise figure across the rest of the band, particularly within the primary operating range of the receiver, remains well-controlled.

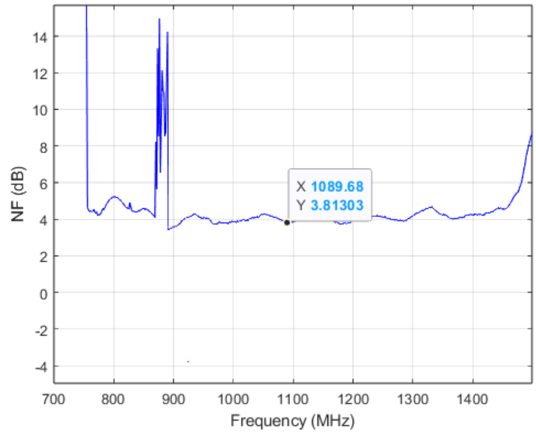

FIGURE 4. Receiver block for DME and ADS-B (870 to 1440MHz) measured NF.

Figures 5, 6, and 7 present the NF measurements for the radios operating at lower frequencies. These results demonstrate consistency across all designed receiver paths, with NF values remaining below 4 dB, validating that the selected RF limiter, LNA, and BPF components maintain low system noise across different radio bands.

Achieving a low and consistent NF is important for avionics applications, where high receiver sensitivity directly impacts communication range, navigation accuracy, and system resilience to interference. In the proposed architecture, the reuse of RF paths among radios with similar frequency characteristics (e.g., VOR and ILS LOC) was intended to minimize size and weight without degrading performance. The consistent NF across all bands confirms that this architectural choice does not introduce additional noise penalties, ensuring good receiver operation across all modes. Furthermore, these NF measurements verify that the RFFE is properly matched and that the signal chain, including protection against high-power signals and selective filtering, was implemented without significantly sacrificing sensitivity. These power spectral plots are used to validate the effectiveness of the bandpass filter in suppressing harmonics. They directly support the system's ability to meet out-of-band emission requirements, which is important for safe operation and certification

TABLE 1. NF of each radio at the radio center frequency.

Radio	Centre Frequency (MHz)	NF
ILS, VOR	118	3.44
COM	128	3.56
GS	332	3.58
ADS-B, TMS, DME	1090	3.81

Table 1 shows the measured NF of the receiver used for all radios, which was used for VOR, ILS, GS, and COM in the

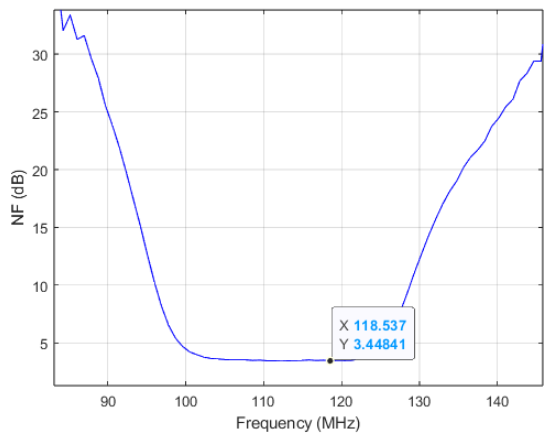

FIGURE 5. Receiver block for VOR and ILS (90MHz to 140MHz) measured NF.

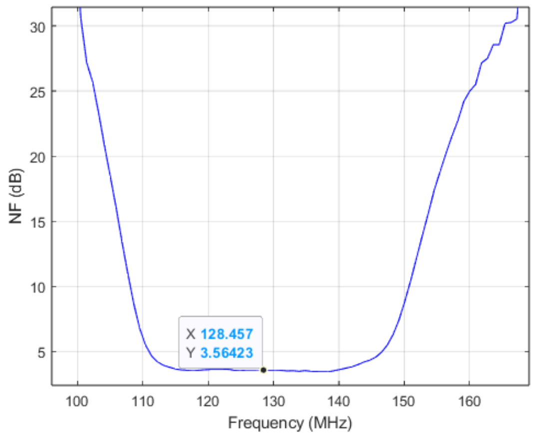

FIGURE 6. Receiver block for COM (100MHz to 160MHz) measured NF.

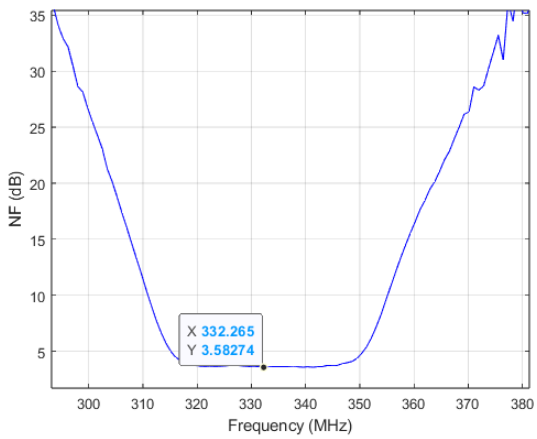

FIGURE 7. Receiver block for GS (300MHz to 370MHz) measured NF.

VHF unit and TMS, ADS-B, and DME in the L-band box. As can be seen, the range is almost the same for all, which suggests the consistency of the design.

C. SPECTRUM UTILIZATION

The importance of not transmitting out-of-band signals cannot be overstated in aviation, particularly for critical systems

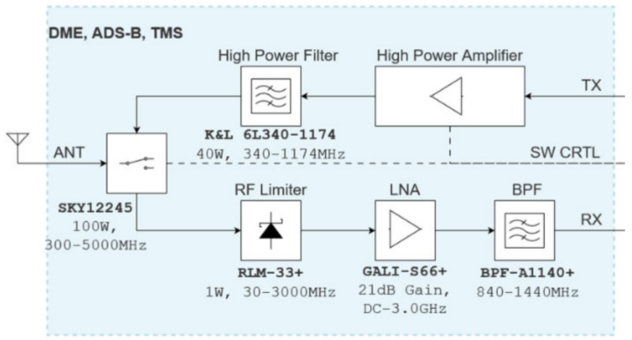

FIGURE 8. L-Band unit with BPF added.

such as Automatic ADS-B, DME, and TMS. Out-of-band emissions, often caused by the non-linearity of high-power transmitters, can lead to interference with adjacent channels and other vital communication and navigation systems, compromising overall safety and operational efficiency. To mitigate these issues, a high-power bandpass filter was proposed at the high-power amplifier (HPA) output to eliminate the 2nd and 3rd harmonics. Figure 8 demonstrates both RX and TX of the L-band block. Specifically, a K&L 6L340-1174 40W bandpass filter is employed, which operates within the 340-1174 MHz range and can support short bursts of up to 100W signals, matching the bandwidth requirements of our target radios. This filter effectively suppresses out-of-band emissions, only transmitting the desired frequency signals. To determine the acceptable duration for 100 W burst signals on a 40 W continuous power-tolerant band pass filter, the following relationship is used:

$$T_{\text{burst}} = \frac{P_{\text{cont}} \cdot T}{P_{\text{burst}} - P_{\text{cont}}} \tag{1}$$

Substitute $P_{\text{cont}} = 40\,\text{W}$, $P_{\text{burst}} = 100\,\text{W}$ into the formula:

$$T_{\text{burst}} = \frac{40 \cdot T}{100 - 40} \tag{2}$$

Simplify the equation:

$$T_{\text{burst}} = \frac{2}{3} T \tag{3}$$

Figure 9 shows the PA output without any filter in the output. This measurement was conducted using a 0 dBm continuous-wave 1090 MHz single-tone signal input to the PA, with a 60 dB attenuator placed before the spectrum analyzer for protection.

The impact of the filter on the transmitted output is illustrated in Figure 10, which demonstrates a significant reduction of the second harmonic at 2180 MHz, achieving a suppression of 48.4 dBc relative to the primary signal, which is a substantial improvement compared to 8.3 dBc in the unfiltered output. The filter introduces less than 0.7 dB insertion loss on the fundamental signal, while effectively suppressing the second and third harmonics by more than 48 dBc. This minimal impact on the desired signal, combined with strong harmonic rejection, allows the transmitter to maintain both high linearity and spectral purity. In aviation

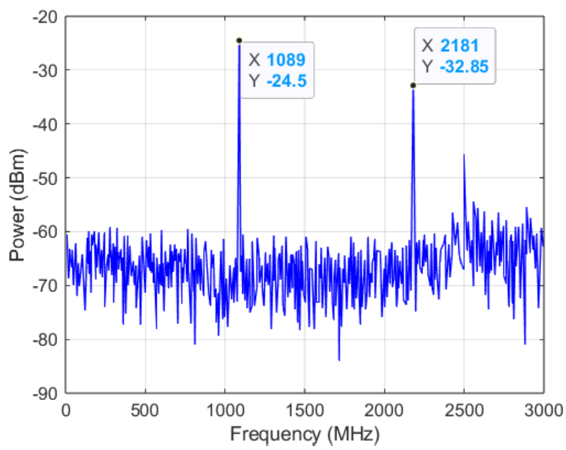

FIGURE 9. PA output signal without BPF.

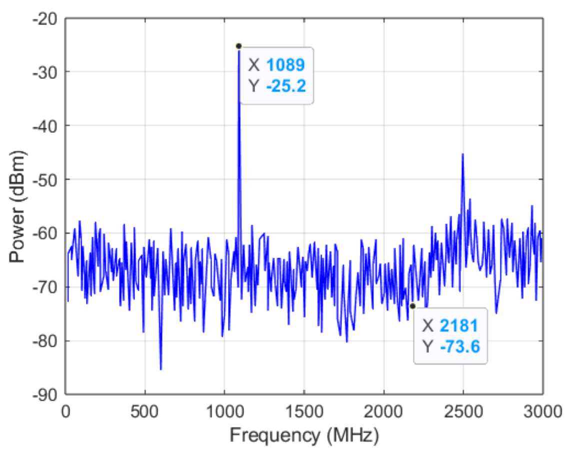

FIGURE 10. PA output signal with BPF.

systems, where multiple radios operate closely together, controlling out-of-band emissions prevents interference with adjacent channels and ensures regulatory compliance. The measured spectral results show that the proposed architecture achieves the necessary signal integrity to support safe and reliable multi-band operation without significantly degrading system efficiency or transmission power.

D. RF SWITCH PROTECTION

Transmitting high power to a solid-state RF switch while it is in RX mode poses a risk of permanent damage to the switch and disruption of the entire RF communication system. Solid-state RF switches are designed to handle low signal levels in RX mode and cannot safely withstand the high RF energy present during transmission. If the transmitter is activated while the switch is not correctly positioned in TX mode, significant power, up to 50W in our system, can leak through the passive isolation path or even dissipate within the internal structures of the switch, such as FETs or PIN diodes, causing failure.

To eliminate the risk of damaging the RF switch or the receiver path, a dedicated logic circuit was implemented to control the sequencing between the RF switch and the PA.

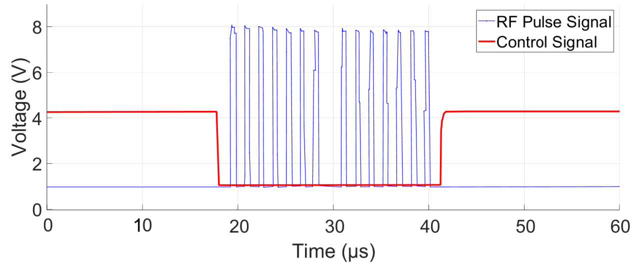

FIGURE 11. RF switch signal timing.

This circuit ensures that the PA remains disabled until the RF switch is fully set to TX mode, avoiding any overlap where high-power transmission could leak into the RX path. The circuit operates by monitoring the PA disable signal and allowing the SDR's control signal to reach the RF switch only after confirming that the PA is safely turned off. As shown in Figure 11, the control signal transitions the RF switch to TX mode one microsecond before the RF pulse transmission begins and returns it to RX mode one microsecond after transmission ends. The timing delays introduced by the logic gates are on the order of a few nanoseconds, which is negligible compared to the switching time and has no impact on system performance.

This protection mechanism is important in multi-radio SDR systems where one RFFE is shared across multiple radio chains, requiring rapid switching without compromising hardware integrity. The RF switch used in our system (SKY12245-492LF) provides approximately 46dB of isolation at 1090MHz. While this limits leakage to about +1dBm when the PA is active, the isolation alone is not a safeguard against destructive failure. Thus, the logic circuit guarantees safe conditions through enforced timing.

Figure 12 and Figure 13 illustrate the functional block diagram and the hardware schematic of the protection logic.

III. FLIGHT TEST

To verify the real-world applicability and practical robustness of the enhanced RFFE architecture described in Section II, we conducted comprehensive flight tests. These tests represent an important progression from controlled laboratory environments to in-flight conditions.

A. FLIGHT TEST SETUP AND PROCEDURE

This segment provides an overview of the flight test setup and procedures, outlining various flight phases during which distinct test cards and maneuvers will be performed. Each phase specifies the altitude and speed necessary for the respective test card. The pilot retains the authority to decline or propose alternative parameters based on area restrictions, weather conditions, and aircraft capabilities. Figure 14 shows the SDR setup inside the aircraft before the flight test, and Figure 15 demonstrates the setup that was prepared for the ground test.

The system is designed to operate with commercially available and standard aircraft antennas that support CNS

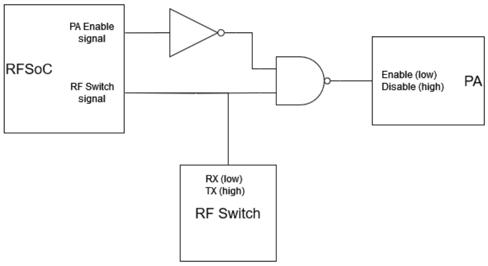

FIGURE 12. RFFE architecture-control signal protection logic.

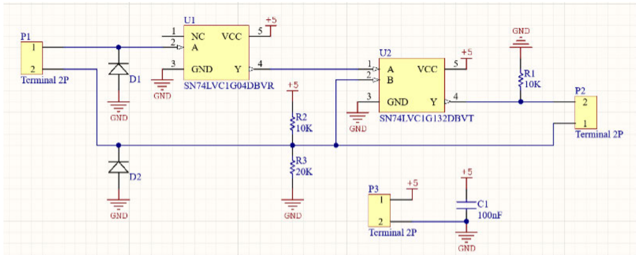

FIGURE 13. Schematic of the protection circuit.

systems. A Trivec-Avant Corp. VOR/LOC/GS combined navigation antenna is used for VOR, LOC, and GS functions, facilitating both navigation and descent guidance. For communication, a Comant CI-122 VHF COM whip antenna enabled reliable two-way radio communication. The Comant CI-105 multi-function antenna supported DME, TMS, and ADS-B functionalities.

B. GROUND TEST

Conducting Electromagnetic Interference (EMI) tests as the first step of flight testing for a transmitter with an output power of up to 50W is crucial for ensuring the safety and reliability of the aircraft's systems. The 50W output power corresponds to the maximum transmission capability of the 1189 - BBM3K5KKO power amplifier integrated into the SDR system. This level was selected to represent a worst-case EMI scenario during pre-flight ground testing. The amplifier is a commercially available L-band module rated for up to 50W continuous output, providing sufficient margin for high-power operation and compliance validation. This initial testing phase is designed to identify and mitigate any potential interference the transmitter may cause to other critical onboard systems, such as navigation, communication, and control systems. By performing EMI tests early, engineers can detect and address any compatibility issues, ensuring that all electronic components can operate harmoniously without causing or suffering from disruptive interference. Before proceeding with the flight test, a comprehensive ground test was conducted to ensure the proper functioning of various systems and components. The ground test involved monitoring reception and transmission audio signal quality across different frequency ranges for VHF communication systems, including L-band frequency. All tests for these frequencies passed with reception and transmission qualities meeting the required 4/5 rating. Additionally,

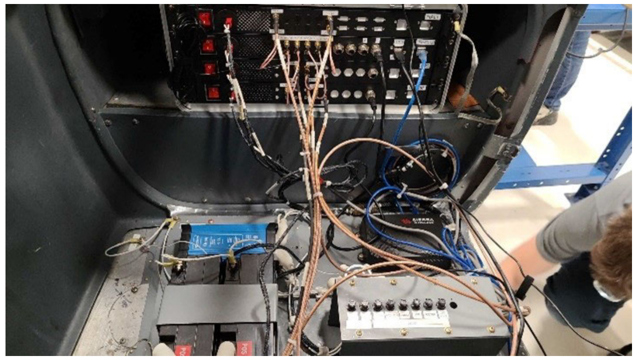

FIGURE 14. SDR and RFFE flight test setup.

FIGURE 15. Flight test ground station.

various operational checks were performed under normal conditions, including the ATC 1 XPDR in ALT mode, cockpit lighting, anti-collision lights, electrical system monitoring, standby magnetic compass, and engine instruments. These components also passed their respective tests, indicating no abnormalities or deviations during DME transmission and prototype operation. The successful results of these ground tests ensure that the systems are functioning correctly and are ready for the subsequent flight tests.

C. FLIGHT TEST RESULTS

The flight test trajectory shown in Figure 16 evaluates the performance of multiple radios integrated into the SDR under actual aviation conditions, validating its RFFE architecture. Figure 17 shows the radios' active slots. The following sections present results on signal power and comparisons with reference data. GGPS data is selected as the comparison baseline due to its high accuracy, widespread use, and established reliability in aviation systems. In this study, it is also used to compute reference values such as the Difference in Depth of Modulation (DDM), which is derived from the aircraft's position relative to the ideal flight path and represents the expected signal deviation a properly aligned radio system should observe. Since verified avionics performance data is often not accessible, especially in light aircraft with limited digital instrumentation, the use of GPS provides a rigorous and standardized method for evaluating system performance in flight environments.

Figures 18, 20, and 22 provide a broad overview of the flight profile, illustrating altitude, distance, and timing data during the test flights. In addition to DDM and positional accuracy, these figures include signal power plots to demonstrate the stability and variation in received signal strength during different flight phases. This helps validate the system's robustness in handling real-time signal dynamics under actual flight conditions, such as touch-and-go maneuvers. These figures contextualize how different maneuvers and flight phases influence the captured data. Figures 19, 21, and 23 zoom into specific flight segments to allow direct comparison of SDR measurements with GPS reference data. The inclusion of signal power in these zoomed views further highlights the SDR system's ability to track and respond to transient fluctuations in complex scenarios.

D. ILS LOC AND GS

In the recent testing of the ILS and GS functionality of the SDR system, a touch-and-go approach was employed during actual flight tests. This method involved an aircraft performing landing approaches that descended to within 50 feet of the runway before ascending again. Due to the directional nature of the antennas used in the system, performance could only be accurately tested during the descent phase until a few seconds at the lowest altitude when the aircraft was aligned with the ILS and GS signals. The data collected a few seconds before (for GS) and after the aircraft began its ascent (the "go" phase) was not helpful in evaluating the system's performance, as the antennas were no longer optimally oriented to receive the signals. The ILS maintains accurate lateral guidance even moments after the aircraft begins to ascend, thanks to stable antenna alignment with the runway's lateral signals. In contrast, the GS experiences reduced accuracy before the aircraft reaches its lowest altitude. This is due to the GS signals being more sensitive to altitude and the directional antennas' susceptibility to signal loss as the aircraft ascends, highlighting challenges in maintaining precise vertical guidance.

By executing multiple touch-and-go maneuvers, we ensured that the SDR system's ability to interpret and respond to ILS and GS signals accurately was thoroughly and reliably assessed. The first figure presents the data collected during a flight maneuver involving three touch-and-go landings.

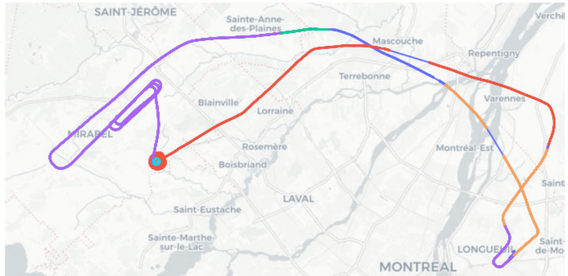

FIGURE 16. Flight trajectory with the active radio identified by trajectory color.

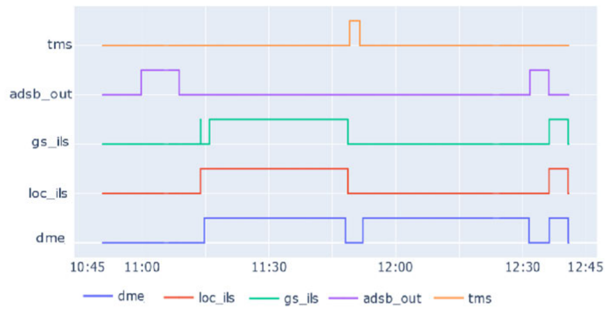

FIGURE 17. Active radio vs time.

Figure 18 depicts the DDM measured by the GPS-based localizer signal, with distinct variations corresponding to the moments of touch-and-go. The middle plot illustrates the altitude and distance over time, where the cyclic pattern reflects the aircraft's descent and ascent during each touch-and-go operation. The bottom plot displays the received and perceived power levels, which remain relatively stable, with noticeable variations coinciding with the maneuvers.

Figure 19 provides a zoomed-in view of one of the touch-and-go events, facilitating a detailed comparison between the GPS-based localizer and the SDR measurements. In the top plot, the DDM from the SDR (denoted as "sari__loc__ddm") is compared to the GPS data, with an additional overlay representing the uncertainty from the Minimum Operational Performance Standards (MOPS) GPS model. The SDR measurements show a high degree of alignment with the GPS data, demonstrating the effectiveness of the SDR in capturing the localizer signal accurately. The bottom plot's power data reveals transient fluctuations aligned with the in-flight maneuver transitions, offering insights into the system's sensitivity and continuity of reception even as antenna alignment shifts. This visualization supports the SDR's real-time responsiveness and ability to maintain the link under changing orientation and motion.

The close match between the SDR and GPS data, with minimal deviation, underscores the SDR's capability to provide reliable localization information that is consistent with established GPS measurements. The middle plot continues to depict the altitude and distance, offering a more focused perspective on the single event. The bottom plot

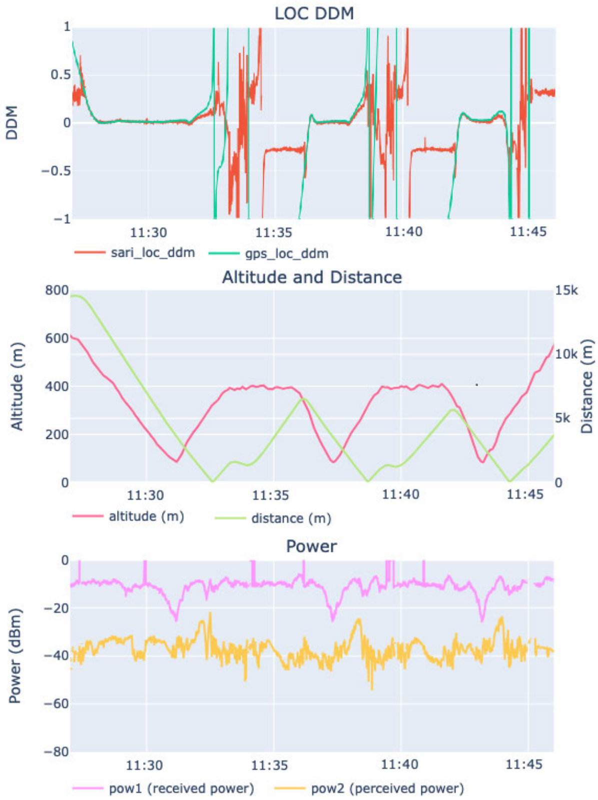

FIGURE 18. Data from three touch-and-go maneuvers for the localizer signal, compared to GPS data.

FIGURE 19. Zoomed data the localizer in comparison to GPS data in the last maneuver.

similarly zooms in on the power measurements, highlighting fluctuations during this specific maneuver. This concentrated view enables a more precise comparison of the performance between the GPS and SDR systems during a single touch-and-go operation, confirming the SDR's robustness in mirroring the GPS-derived DDM values. The comparison shows a perfect fit of GS location estimation with the GPS measurement, considering that the GS is not meant to maintain precision as the aircraft gets closer to the transmitter.

Figure 20 presents the data collected for the GS signal during a flight maneuver involving three touch-and-go landings. The top plot displays the difference in DDM measured by the GPS-based GS signal and the corresponding DDM obtained using the SDR. The SDR measurements, denoted as "sari__gs__ddm," exhibit significant variations during the touch-and-go landings, closely aligning with the GPS-based measurements ("gps__gs__ddm"). The middle plot illustrates the altitude and distance over time, where the altitude variations correspond to the aircraft's descent and ascent during each touch-and-go, as reflected in the cyclic patterns. The bottom plot shows the received and perceived power levels, which maintain relative stability with some fluctuations during the maneuver.

Figure 21 offers a zoomed-in view of one of the touch-and-go events, providing a detailed comparison between the SDR and GPS measurements for the GS signal. In the top plot, the SDR-derived DDM closely matches the GPS-based DDM, with the addition of an overlay representing the uncertainty from the MOPS GPS model. The SDR's performance demonstrates a strong correlation with the GPS data, accurately capturing the GS signal's characteristics. The middle plot continues to detail the altitude and distance for the selected event, providing a focused perspective that highlights the precision of the measurements. The power trace further reflects minor signal level fluctuations during the maneuver, showing the RFFE SDR system's robustness in capturing consistent data during rapid altitude and angle changes. The bottom plot further zooms in on the power levels, illustrating fluctuations during the specific maneuver. This closer examination reaffirms the SDR's capability to reliably replicate GPS-derived measurements, particularly in capturing the nuanced changes in the GS DDM during a single touch-and-go operation.

Including a diagram of the received and perceived signal levels helps demonstrate that the RFFE consistently maintained appropriate signal amplitude, enabling effective scaling through the automatic gain control (AGC) loop and confirming the front-end's robustness under varying link conditions.

E. DME TEST

A test was conducted to compare DME radio in the SDR system with MOPS GPS, revealing a highly accurate

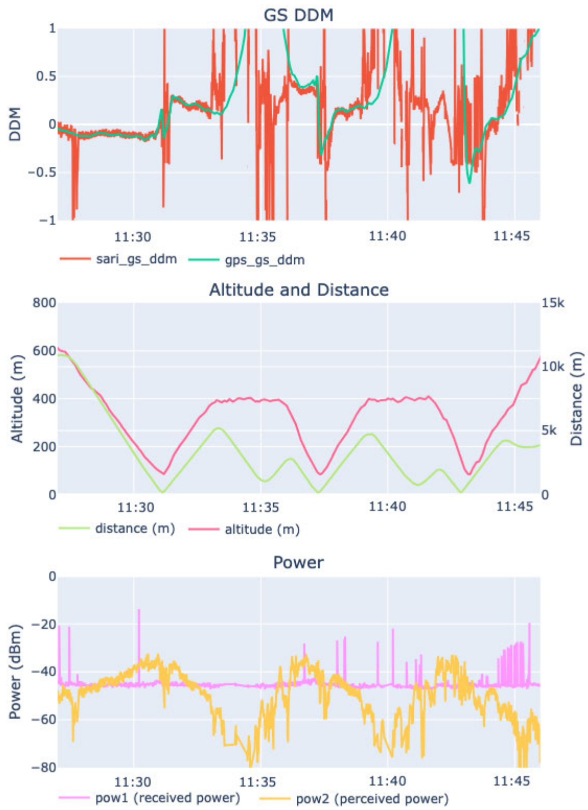

FIGURE 20. **Data from three touch-and-go maneuvers for the GS signal, compared to GPS data.**

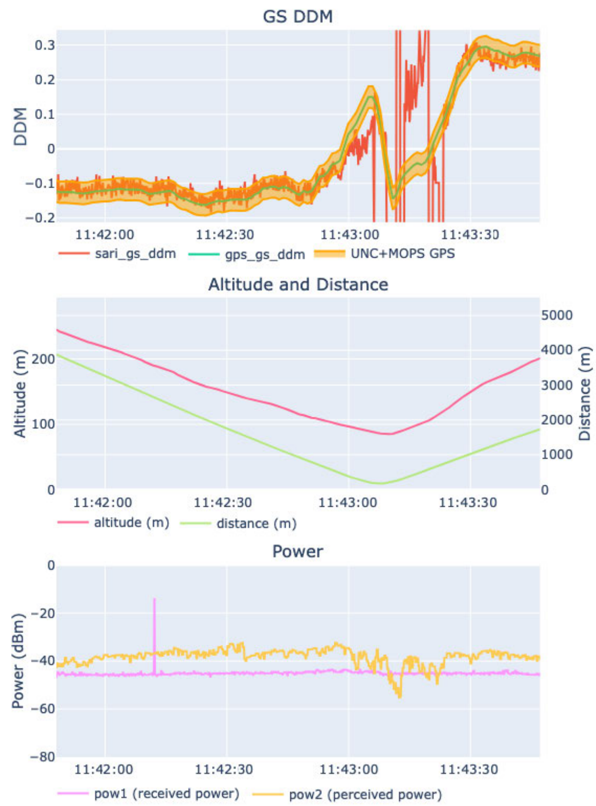

FIGURE 21. **Zoomed data the GS in comparison to GPS data in the last maneuver.**

correlation between the two methods. This test affirmed that both RFFE and SDR functionalities performed as expected.

As shown in Figure 22, as an example, it can be seen that between 12:21 and 12:26, the SDR seamlessly switched to another L-band radio, temporarily interrupting DME data availability. This capability demonstrates the system's efficiency in using a single RFFE for multiple radios, enhancing operational flexibility without compromising overall performance.

Figure 23 displays the furthest distance the aircraft travelled, highlighting the precision of the measurement with exceptional accuracy. This figure illustrates the reliability of the SDR system in tracking and displaying distance measurements. During the test, there were instances where the DME briefly lost signal and entered search mode. Despite these interruptions, the system quickly resumed measurements after several attempts and successfully re-locked onto the position. This capability underscores the robustness of the SDR system in maintaining consistent and reliable distance measurements even under challenging conditions, ensuring accurate navigation information throughout the test duration. These signal power readings, especially in the zoomed-in view, offer evidence of the system's quick recovery from signal dropout, validating the practical agility and resilience of the RFFE-SDR system during transitions between radio modes.

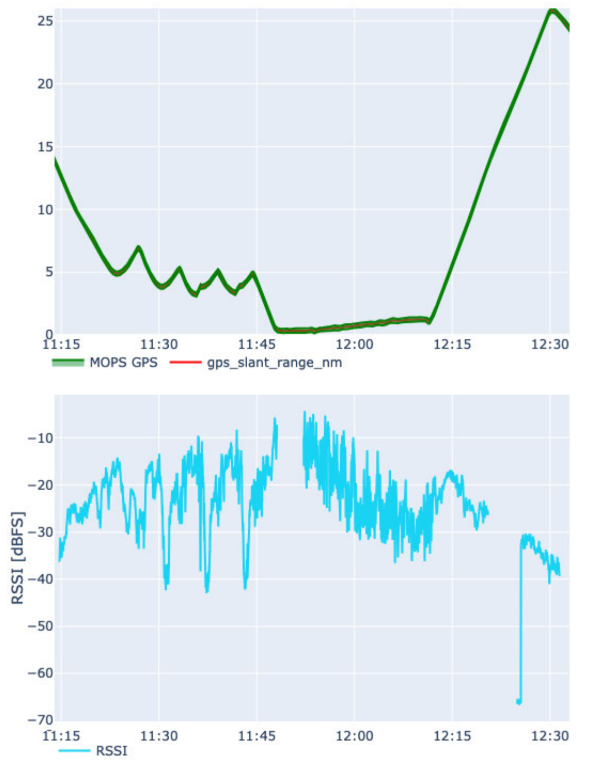

FIGURE 22. **Data from three touch-and-go maneuvers for the DME signal, compared to GPS data.**

A quantitative analysis of the DME performance was also carried out by comparing the SDR-measured slant

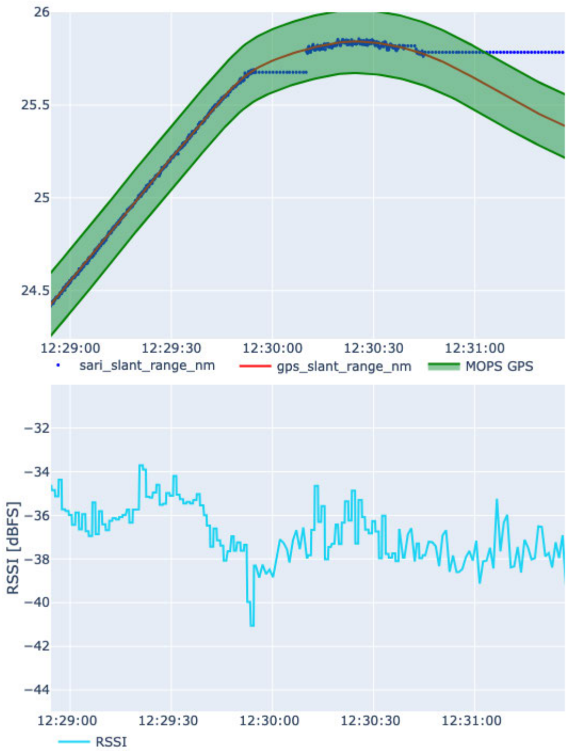

FIGURE 23. Zoomed data the DME in comparison to GPS data in the last maneuver.

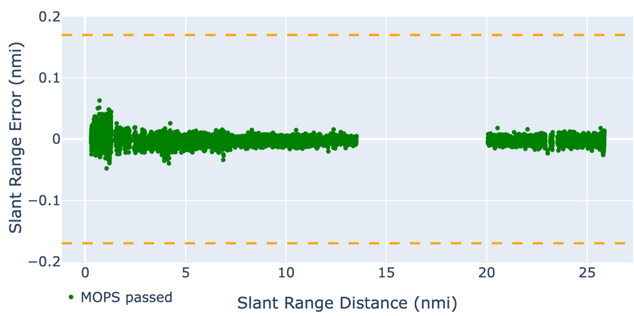

FIGURE 24. DME slant range compared to DO-189 accepted threshold.

range against a GPS-derived reference. The resulting error distribution, shown in Figure 24, confirms that all recorded points remained well within the ±0.17 NM limit defined by DO-189 [19], demonstrating stable and compliant tracking throughout the evaluated distance range. This method follows the benchmarking approach presented in [20], where GPS-based truth data is used to assess the flight performance of SDR-based avionics. For the ILS signals, theoretical DDM values were computed using GPS position and antenna geometry as described in ICAO Annex 10 [21]. Although no statistical error metrics are reported for ILS in this work, visual inspection of Figures 19 and 21 shows strong agreement between the SDR output and GPS-based references during multiple touch-and-go maneuvers. These observations confirm that the SDR implementation adheres to expected signal characteristics under real-world flight conditions.

IV. CONCLUSION

This study presents, for the first time, detailed flight-test validation of an adaptive RFFE architecture for SDR avionics systems, underscoring significant advancements beyond prior laboratory-focused research. The successful in-flight demonstrations validate the architecture's operational effectiveness and potential for future practical aviation deployment.

In conclusion, the new revision of the RFFE has undergone thorough testing and is ready for real-world flight tests. The testing process included three stages: lab, ground, and flight tests, all of which were successfully completed. Key updates were implemented to ensure high signal quality. These updates include improving the LNA in both electrical and mechanical design, adding a filter at the PA output, and integrating a protection circuit to prevent high-power signal transmission while the RF switch is in RX mode. Multiple flight tests demonstrated that the RFFE consistently provided high-quality signals, enabling the SDR to process, detect, and demodulate signals effectively while simultaneously transmitting high-power signals.

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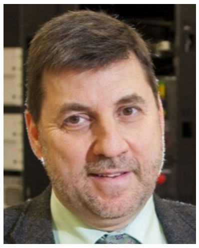

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